**MINI PROJECT**

**MACHINE LEARNING**

**A Mini Project on**

# *Analysis of Forest Loss and Carbon Emissions in India Using Dashboard Visualization*

**BY**


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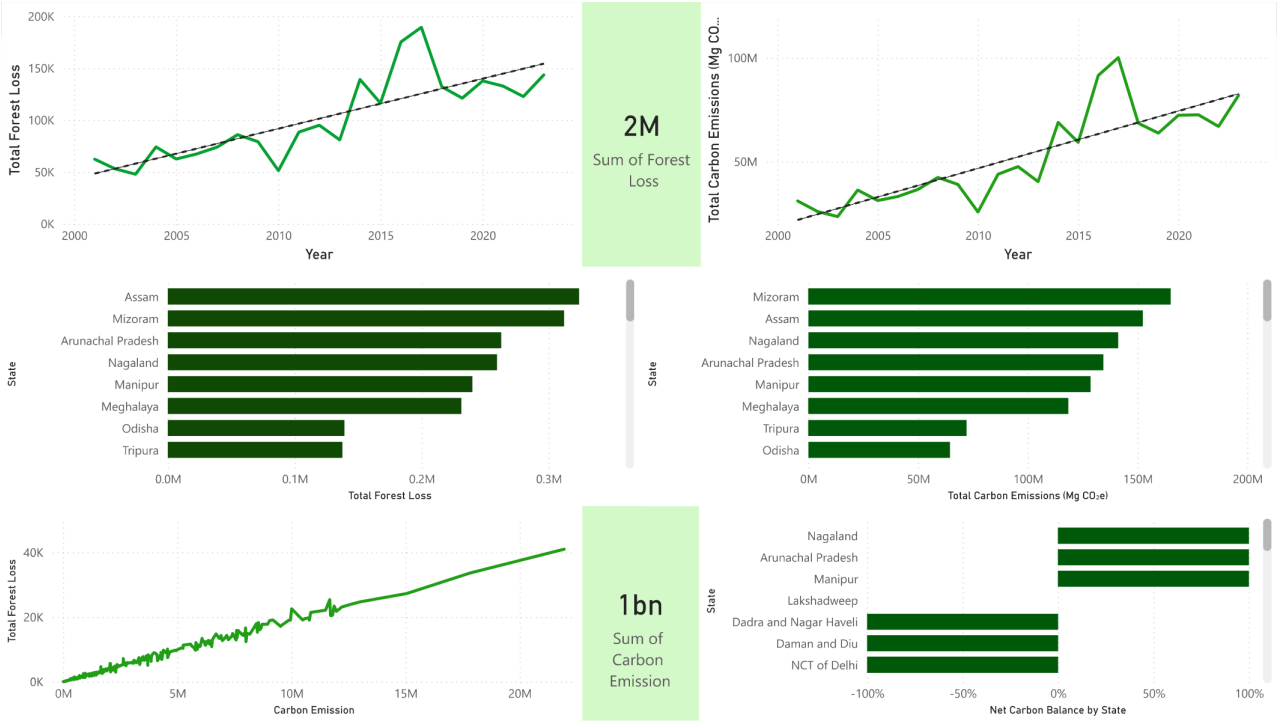


# ABSTRACT

Forests play a vital role in maintaining ecological balance, supporting biodiversity, regulating hydrological cycles, and acting as carbon sinks that absorb atmospheric carbon dioxide. In developing countries like India, increasing population pressure, urban expansion, infrastructure development, agricultural conversion, mining activities, and industrialization have contributed significantly to forest degradation over the past two decades. This study presents a comprehensive analysis of forest loss and associated carbon emissions in India from 2001 to 2023 using satellite-derived datasets obtained from **Global Forest Watch**, an international forest monitoring platform managed by the World Resources Institute.

The research adopts a data-driven and visualization-oriented approach to evaluate long-term trends in forest loss, examine carbon emissions resulting from deforestation,

compare state-wise variations, and assess the relationship between forest loss and net carbon flux. The dataset was processed and transformed using Microsoft Excel and visualized through interactive dashboards in Power BI. Trend analysis, comparative charts, and key performance indicators were used to interpret the results effectively. The findings indicate a steady increase in forest loss over the study period, with a parallel rise in carbon emissions. Northeastern states such as Mizoram, Assam, and Arunachal Pradesh emerged as

significant contributors to forest cover reduction. The study also observes that several regions are transitioning from carbon sinks to carbon sources, as indicated by net flux values. Furthermore, recent updates from Global Forest Watch reveal that India lost 18,200 hectares of primary forest in 2024, suggesting that the issue of forest degradation persists beyond the analyzed timeframe.

The study highlights the urgent need for stronger forest governance, sustainable land-use planning, and policy interventions aligned with climate mitigation goals. The dashboard-based analytical framework provides policymakers, researchers, and environmental planners with an accessible and interpretable decision-support tool.

## KEYWORDS

Forest Loss, Carbon Emissions, Climate Change, India, Data Visualization, Power BI, Global Forest Watch, Sustainability

## OBJECTIVE

The primary objective of this research is to systematically evaluate forest loss and carbon emissions trends in India over a 23-year period (2001–2023) using satellite-based environmental data. The study aims to bridge the gap between raw environmental datasets and actionable insights by applying visualization-driven analytics.

The first objective is to analyze forest loss trends in India from 2001 to 2023. This involves examining annual forest cover reduction patterns to determine whether deforestation is accelerating, stabilizing, or declining over time. Identifying long-term trends allows policymakers to evaluate the effectiveness of conservation measures implemented during the study period.

The second objective is to study carbon emissions related to forest degradation. Forests act as carbon sinks, and their destruction releases stored carbon into the atmosphere in the form of carbon dioxide. By examining emission trends, the study assesses the climate impact of deforestation and its contribution to greenhouse gas accumulation.

The third objective is to compare forest loss and carbon emissions across Indian states. India is geographically and ecologically diverse, and forest loss patterns vary significantly by region. State-wise comparison helps identify high-risk areas, especially biodiversity-rich and ecologically sensitive regions such as the northeastern states.

The fourth objective is to understand the relationship between forest loss and carbon emissions. This involves examining whether increases in forest loss correspond with increases in emissions, thereby establishing a correlation between deforestation and climate impact.

The fifth objective is to present insights using an interactive dashboard. Data visualization enhances comprehension of complex environmental datasets and supports evidence-based policymaking. The dashboard serves as a decision-support tool that enables users to filter data by year and state, observe trends, and derive actionable insights.

## INTRODUCTION

Forests are among the most critical natural resources on Earth. They regulate climate, conserve biodiversity, prevent soil erosion, and support millions of livelihoods. In the context of climate change, forests serve as carbon sinks by absorbing atmospheric carbon dioxide through photosynthesis and storing it in biomass and soil. However, when forests are cleared or degraded, stored carbon is released back into the atmosphere, contributing to global warming.

India, as one of the world's fastest-growing economies, faces significant developmental pressures. Rapid urbanization, infrastructure expansion, agricultural intensification, mining operations, and industrial growth have placed increasing stress on forest ecosystems. Although India has implemented conservation programs such as afforestation initiatives and forest protection laws, deforestation and forest degradation remain major environmental challenges.

The northeastern region of India, known for its rich biodiversity and dense forest cover, has experienced considerable forest loss due to shifting cultivation practices, infrastructure development, and land-use changes. Central Indian states have also faced deforestation linked to mining and industrial activities.

With advancements in satellite technology and geospatial monitoring, it has become possible to track forest cover changes with high accuracy. Platforms such as Global Forest Watch provide open-access, satellite-based data on forest loss, carbon emissions, and net carbon flux. These datasets enable researchers to conduct long-term, large-scale environmental assessments.

This study leverages satellite-derived data to examine forest loss and carbon emissions trends in India from 2001 to 2023. By integrating statistical analysis with dashboard visualization, the research enhances understanding of deforestation patterns and their climate implications. The inclusion of 2024 data showing continued primary forest loss emphasizes the ongoing urgency of forest conservation efforts.

## WORKFLOW

The research workflow followed a structured and systematic approach to ensure data accuracy, analytical clarity, and meaningful visualization outcomes.

The first step involved data collection from Global Forest Watch. The dataset included annual forest loss (in hectares), carbon emissions (in metric tons), and net carbon flux values for India and its states. The data was downloaded in tabular format for further processing.

The second step involved data cleaning and preprocessing. This stage ensured data

reliability by removing duplicate records, checking inconsistencies, handling missing values, and validating year-wise entries. Data integrity is essential to avoid misleading interpretations.

The third step involved data transformation using Microsoft Excel and Power BI. The data was organized year-wise and state-wise. Aggregations were performed to compute total forest loss, total emissions, and average values. Relationships between variables were structured for visualization.

The fourth step involved dashboard design and visualization in Power BI. Interactive visual elements such as line charts, bar graphs, KPI indicators, slicers, and filters were incorporated. The dashboard was designed to allow dynamic exploration of trends over time and across states.

The fifth step involved data analysis and interpretation. Patterns, correlations, and anomalies were examined. Comparative analysis was conducted to identify high-loss states and peak emission years.

The final step involved result documentation. The insights derived from visual analysis were documented in a structured format to support hypothesis testing and policy recommendations.

## METHODS

### Data Source

The primary data source for this study was Global Forest Watch, a satellite-based monitoring system developed by the World Resources Institute. The platform integrates satellite imagery, remote sensing technology, and geospatial datasets to provide reliable and updated information on forest cover changes worldwide.

### Data Processing

Data preprocessing involved multiple stages to ensure analytical consistency.

Duplicate entries were removed to prevent redundancy. Missing values were either interpolated or excluded depending on their frequency and impact. Data was categorized year-wise (2001–2023) and state-wise for comparative analysis. Aggregation techniques were applied to calculate total forest loss and emission values.

Normalization techniques were also considered to compare states with varying geographical areas. This ensured fair comparison across large and small states.

**Tools Used**

Microsoft Excel was used for initial data cleaning, sorting, filtering, and aggregation. Power BI was used for advanced data modeling, relationship mapping, and interactive dashboard development.

**Visualization Techniques**

Line charts were used to analyze time-series trends. Bar charts were used for state-wise comparisons. KPI cards displayed total forest loss, total carbon emissions, and net carbon flux values. Scatter plots were used to visually assess the correlation between forest loss and emissions.

## HYPOTHESIS TESTING

Null Hypothesis ($H_0$):

There is no significant relationship between forest loss and carbon emissions in India.

Alternative Hypothesis ($H_1$):

There is a significant relationship between forest loss and carbon emissions in India.

Method:

To examine the relationship between forest loss and carbon emissions, a visual trend analysis was conducted using **line graphs in Power BI**. Two line graphs were created

— one representing annual forest loss (in hectares) and another representing annual carbon emissions over the period 2001 to 2023. These graphs were plotted year-wise to observe patterns and variations over time.

The purpose of using line graphs was to analyze whether both variables follow a similar upward or downward trend during the same time periods. By comparing the direction and movement of the lines across years, it becomes possible to visually identify whether increases in forest loss correspond with increases in carbon emissions.

Observation:
The line graphs show that during years when forest loss increased, carbon emissions also showed a corresponding rise. Similarly, in years where forest loss slightly declined or stabilized, carbon emissions reflected a similar pattern. The parallel movement of both lines across the timeline indicates that the two variables are positively associated.

The consistent upward trend observed in both forest loss and carbon emissions over the study period suggests that deforestation activities are contributing directly to increased carbon release into the atmosphere.

Decision:
Based on the visual trend analysis using line graphs, the null hypothesis ($H_0$) is rejected. The alternative hypothesis ($H_1$) is accepted, indicating that there is a significant positive relationship between forest loss and carbon emissions in India.

## RESULTS

The analysis of data from 2001 to 2023 reveals several important findings regarding forest loss, carbon emissions, and net carbon flux in India.

1. Increasing Trend in Forest Loss
   The line graph representing forest loss shows an overall increasing trend during the study period. Although minor fluctuations are observed in certain years, the general direction of the graph moves upward. This indicates that forest degradation has gradually intensified over the last two decades. The continuous rise suggests that conservation measures have not been sufficient to fully offset

deforestation pressures.

2. High Forest Loss in Northeastern States
   State-wise analysis indicates that northeastern states such as Mizoram, Assam, and Arunachal Pradesh recorded comparatively higher forest loss. These states have large forest cover areas, and changes in land use, shifting cultivation, and development activities may have contributed to this trend. The data highlights that forest degradation is not uniform across India and varies significantly by region.

3. Increase in Carbon Emissions
   The line graph for carbon emissions shows a pattern similar to forest loss. Over time, carbon emissions increased steadily, especially during years with higher forest loss values. Since forests act as carbon sinks, their destruction results in the release of stored carbon into the atmosphere. The increasing emission trend indicates that forest degradation is contributing to higher greenhouse gas concentrations.

4. Positive Relationship Between Forest Loss and Carbon Emissions
   When comparing the two line graphs, it is evident that both variables follow a similar upward movement across the years. This parallel pattern confirms a strong positive relationship between forest loss and carbon emissions. The visual alignment of trends strengthens the conclusion that deforestation significantly influences emission levels.

5. Net Carbon Flux Patterns
   Net carbon flux values indicate that several regions are gradually shifting from being carbon sinks to becoming carbon sources. A carbon sink absorbs more carbon than it emits, whereas a carbon source releases more carbon than it absorbs. The data suggests that increasing forest loss is weakening the carbon absorption capacity of many regions in India.

Overall, the dashboard effectively presents these patterns using clear visualizations, making it easier to interpret complex environmental data and identify long-term trends.

## CONCLUSION

This study concludes that forest loss in India has shown a significant increasing trend from 2001 to 2023. The analysis clearly indicates that deforestation remains a persistent environmental issue despite various conservation efforts. The upward movement in forest loss over the years demonstrates the growing pressure on natural ecosystems due to urbanization, industrialization, and land-use changes.

The study also confirms that carbon emissions have increased alongside forest degradation. Since forests store large amounts of carbon, their destruction directly contributes to the release of greenhouse gases into the atmosphere. The parallel upward trends in forest loss and carbon emissions highlight the environmental impact of deforestation on climate change.

The hypothesis testing using line graph analysis supports the existence of a significant positive relationship between forest loss and carbon emissions. This means that as forest loss increases, carbon emissions also tend to increase. Therefore, forest conservation plays a crucial role in climate change mitigation.

Additionally, the analysis of net carbon flux indicates that many regions are losing their ability to function as effective carbon sinks. This shift poses a serious threat to environmental sustainability and national climate goals.

The dashboard-based approach used in this study proves to be an effective method for analyzing and presenting environmental data. It simplifies complex datasets and helps policymakers, researchers, and environmental authorities make informed decisions.

In conclusion, strong conservation measures, sustainable forest management practices, and effective monitoring systems are essential to reduce future forest loss and carbon emissions in India. Protecting forests is not only important for biodiversity but also critical for long-term climate stability and sustainable development.